Software Engineering
Assignment-2
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following statements regarding milestones in a software life cycle model are correct?

a. Milestones help project managers track the progress of a software project
b. Entry into and exit from a development phase are considered important milestones
c. Absence of proper milestones always results in poor software quality
d. Milestones help in monitoring and controlling project activities
e. Milestones eliminate the need for project documentation and reviews

Correct Answer: a. Milestones help project managers track the progress of a software project
b. Entry into and exit from a development phase are considered important milestones
d. Milestones help in monitoring and controlling project activities

Detailed Solution:

Milestones serve as measurable checkpoints that help project managers track progress, monitor activities, and mark phase transitions (entry/exit). However, their absence does not always lead to poor quality, and they do not eliminate the need for documentation and reviews—those remain essential regardless of milestones.

QUESTION 2:

A student while working on a software project, is suffering from the "99% complete syndrome. Which of the following situations are likely indicators of this syndrome?

a. The student repeatedly claims the project is almost finished, but important tasks still remain incomplete.
b. Most modules are coded, but testing and debugging have not been properly completed.
c. The student accurately estimates the remaining work and delivers the project on time.
d. Documentation, integration, and validation activities are postponed until the end.
e. The project progress appears high on paper, but the final deliverable is not ready for deployment.

Correct Answer: a. The student repeatedly claims the project is almost finished, but important tasks still remain incomplete.
b. Most modules are coded, but testing and debugging have not been properly completed.
d. Documentation, integration, and validation activities are postponed until the end.
e. The project progress appears high on paper, but the final deliverable is not ready for deployment.

Detailed Solution:

The "99% complete syndrome" is marked by repeated claims of near-completion while critical tasks like testing, debugging, integration, and documentation remain unfinished—progress looks high on paper but the software is not deployable. Accurate estimation and on-time delivery (option c) are the opposite of this syndrome, so they are not indicators.

QUESTION 3:

Which of the following are important factors to be assessed to determine the technical feasibility of a software project?

a. Availability of required hardware and software resources
b. Technical expertise of the development team
c. Compatibility with existing systems
d. Expected profit from the software product
e. Ability of the technology to handle expected system performance requirements
f. Estimated market demand for the product

Correct Answer: a. Availability of required hardware and software resources
b. Technical expertise of the development team
c. Compatibility with existing systems
e. Ability of the technology to handle expected system performance requirements

Detailed Solution:

Technical feasibility assesses whether the project can be built with available resources, team expertise, system compatibility, and performance capabilities—all technical concerns. Expected profit (d) and market demand (f) relate to economic and market feasibility, not technical feasibility.

QUESTION 4:

Which of the following statements about the Classical Waterfall Model are correct?

a. The development process progresses through a sequence of distinct phases.
b. Requirements are expected to be completely specified before development begins.
c. The model supports frequent backtracking between all phases during normal execution.
d. Testing is generally performed after the coding phase is completed.
e. The Waterfall Model emphasizes continuous customer feedback throughout development.

Correct Answer: a. The development process progresses through a sequence of distinct phases.

b. Requirements are expected to be completely specified before development begins.

d. Testing is generally performed after the coding phase is completed

Detailed Solution:

The Classical Waterfall Model follows a linear, sequential progression through distinct phases, with complete requirements specified upfront and testing performed after coding. It does not support frequent backtracking (c) nor emphasize continuous customer feedback (e)—those are characteristics of iterative or Agile models.

QUESTION 5:

Which of the following statements regarding software maintenance are correct?

a. Corrective maintenance involves fixing errors that were not detected during software development.
b. Corrective maintenance aims at addressing system defects.
c. Adaptive maintenance is performed to port software to a new environment or platform.
d. Corrective maintenance mainly focuses on adding new features requested by users.
e. Adaptive maintenance may be required when migrating software to a new operating system.
f. Perfective maintenance can improve the efficiency and implementation of a system.

Correct Answer:

a. Corrective maintenance involves fixing errors that were not detected during software development.
b. Corrective maintenance aims at addressing system defects.
c. Adaptive maintenance is performed to port software to a new environment or platform.
e. Adaptive maintenance may be required when migrating software to a new operating system.
f. Perfective maintenance can improve the efficiency and implementation of a system.

Detailed Solution:

Corrective maintenance fixes bugs/defects (options a and b), adaptive maintenance handles environment changes like new OS or platforms (c and e), and perfective maintenance improves efficiency or implementation (f). Adding new features (d) is actually perfective maintenance, not corrective, so option d is incorrect.

QUESTION 6:

Which of the following statements regarding phase containment of errors in software engineering are correct?

a. Errors should ideally be detected and corrected in the same phase in which they are introduced.
b. The cost of correcting an error generally increases if it is detected in later phases of development.
c. Phase containment of errors helps reduce software development and maintenance costs.
d. Errors introduced during requirements analysis are easiest to fix after system deployment.
e. Testing is the only phase in which software errors can be detected.

Correct Answer:
a. Errors should ideally be detected and corrected in the same phase in which they are introduced.
b. The cost of correcting an error generally increases if it is detected in later phases of development.
c. Phase containment of errors helps reduce software development and maintenance costs.

Detailed Solution:

Phase containment emphasizes detecting and fixing errors in the same phase they are introduced, which reduces costly rework—fixing errors later (especially after deployment) is far more expensive. Errors can be detected through reviews and inspections in any phase, not just testing (e), and requirements errors are hardest to fix after deployment, not easiest (d).

QUESTION 7:

Which of the following are commonly cited limitations of the classical waterfall model in modern software development projects?

a. Difficulty in accommodating changing requirements after development begins
b. Requirement for a complete and stable requirements specification at an early stage
c. Delayed delivery of working software until the end of the project
d. Continuous customer feedback available throughout all development phases
e. Limited flexibility in handling evolving user needs
f. Frequent incremental releases delivered during development

Correct Answer:
a. Difficulty in accommodating changing requirements after development begins
b. Requirement for a complete and stable requirements specification at an early stage
c. Delayed delivery of working software until the end of the project
e. Limited flexibility in handling evolving user needs

Detailed Solution:

The classical waterfall model struggles with changing requirements, requires fully stable specifications upfront, delivers working software only at the end, and offers little flexibility for evolving user needs—these are its key limitations. Continuous customer feedback (d) and frequent incremental releases (f) are not features of waterfall; they describe iterative/Agile models.

QUESTION 8:

Which of the following statements about the V-model of software development are correct?

a. The V-model emphasizes verification and validation activities.
b. Each development phase is associated with a corresponding testing phase.
c. Coding is performed before system design begins.
d. Test planning begins early in the development life cycle.
e. The V-model is well-suited for projects with stable and clearly defined requirements.
f. The V-model completely eliminates the need for software testing.

Correct Answer:

a. The V-model emphasizes verification and validation activities.
b. Each development phase is associated with a corresponding testing phase.
d. Test planning begins early in the development life cycle.
e. The V-model is well-suited for projects with stable and clearly defined requirements.

Detailed Solution:

The V-model emphasizes verification and validation, pairs each development phase with a corresponding testing phase, starts test planning early, and is well-suited for projects with stable, clearly defined requirements. However, coding is performed after system design (not before), and the model does not eliminate testing—it actually stresses its importance.

QUESTION 9:

The prototyping software development model is most suitable for projects facing which of the following situations?

a. Requirements are unclear or likely to change frequently
b. Users are unable to clearly specify their expectations at the beginning
c. Early user feedback is required to refine the system requirements
d. The project requirements are completely stable and well defined
e. Developers need to quickly demonstrate a working model of the system
f. The project mainly focuses on minimizing hardware manufacturing costs

Correct Answer: a. Requirements are unclear or likely to change frequently
b. Users are unable to clearly specify their expectations at the beginning
c. Early user feedback is required to refine the system requirements
e. Developers need to quickly demonstrate a working model of the system

Detailed Solution:

The prototyping model is ideal when requirements are unclear, users cannot specify expectations upfront, early feedback is needed to refine requirements, or a working model must be demonstrated quickly. It is not suitable for stable, well-defined requirements (d), and it addresses software uncertainty, not hardware cost minimization (f).

QUESTION 10:

Which of the following can be considered as project milestones in a software life cycle?

a. Requirements specification approved
b. Coding standards defined
c. System design completed
d. Use of modular programming
e. User acceptance testing completed
f. Final software delivery completed

Correct Answer: a. Requirements specification approved
c. System design completed
e. User acceptance testing completed
f. Final software delivery completed

Detailed Solution:

Project milestones are significant, measurable events like approval of requirements, completion of system design, user acceptance testing, and final delivery—these mark major progress points. Defining coding standards (b) and using modular programming (d) are development practices or guidelines, not major project milestones.

******THE END******